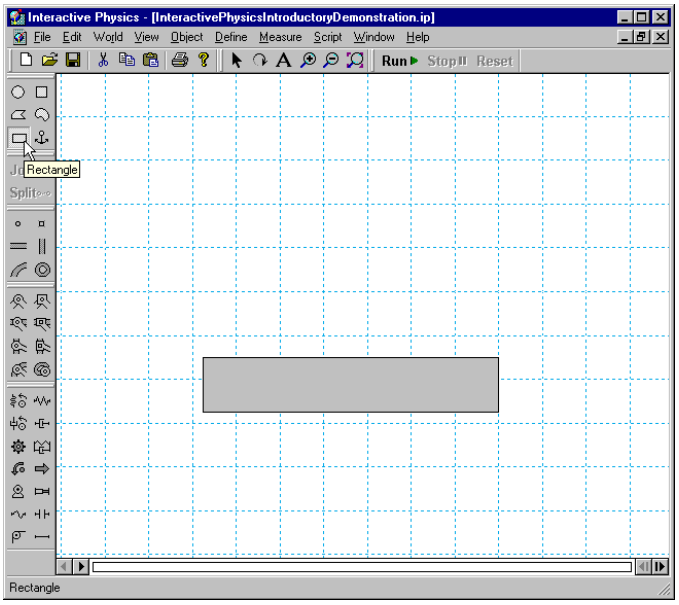
1.1 Starting Interactive Physics

1. Ensure that Interactive Physics is installed on your computer.
2. From the Start Menu, click on Programs and then Interactive Physics 2000. This opens a new Interactive Physics document.



1.2 Creating a Falling Block

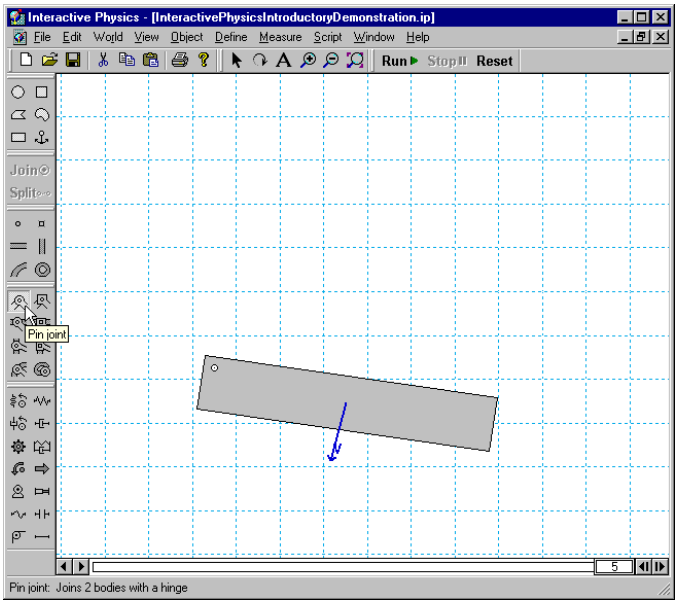
1. The first simulation is Newton's first experiment, dropping a block.
2. To draw a rectangle, click on the Rectangle tool, then click in the workspace and draw a long thin rectangular block.
3. To run the simulation and see the block fall due to gravity, click Run.
4. Click Reset to reset the simulation.

1.3 Adding a Velocity Vector

1. To add a velocity vector, click on the rectangle.
2. From the Define menu, click on Vectors and then Velocity.
3. Click Run and observe that the vector changes magnitude as the block falls.
4. Click Reset.

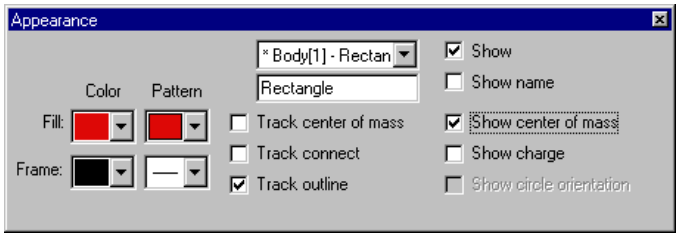
1.4 Making a Pendulum

1. To make a pendulum, click on the Pin joint tool and then click on the upper left-hand corner of the rectangle.
2. Click Run and observe the pendulum's motion.
3. Click Reset.



1.5 Changing an Object's Appearance

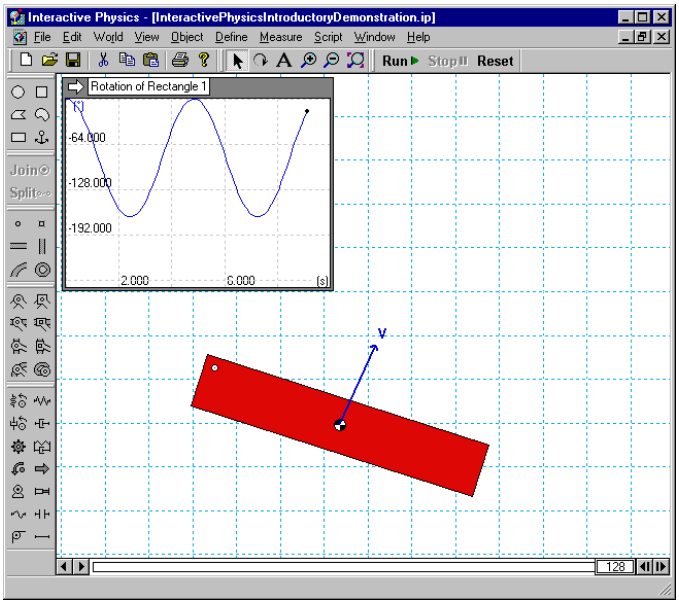
1. To change the appearance of the rectangle, select it (click on it once), then go under the Windows menu and select Appearance. Change the fill color and click the box titled "Show center of mass."



2. Close the Appearance window and run the simulation again. Notice that changing the rectangle's appearance does not affect its motion. Click Reset.

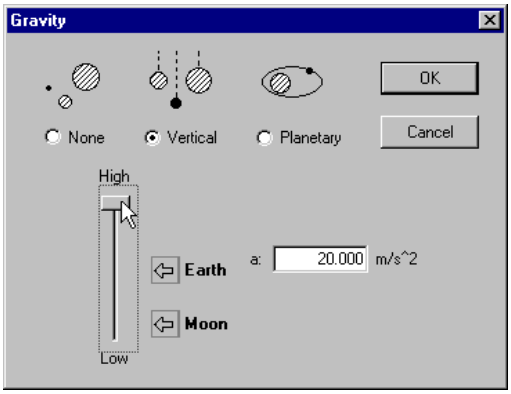
1.6 Graphing the Pendulum's Motion

1. To graph the pendulum's motion, select the rectangle. Under the Measure menu, select Position, then select Rotation Graph.
2. To collect data, click Run, the data can be displayed as a graph, a bar chart, or a number. (Notice that the data display can be changed while running the simulation.) Click Reset.
3. From the graph, the amplitude and frequency of the pendulum's motion can be determined.
4. To make the graph larger, click on the graph and drag its lower right-hand corner to the right.



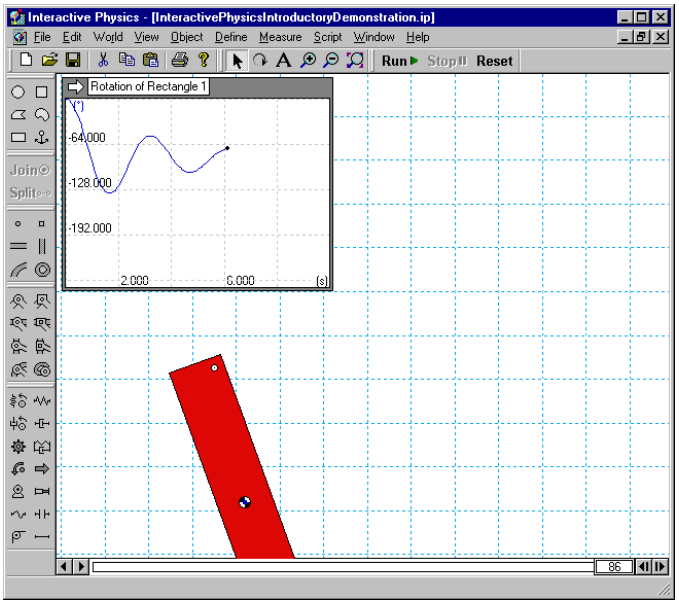
1.7 Changing Gravity

1. To change gravity, click on the World menu, select Gravity, slide the slider to the top for the value 20 m/sec^2, and click [OK].
2. Click Run and observe that, in agreement with theoretical and experimental predictions, the pendulum has a higher natural frequency.



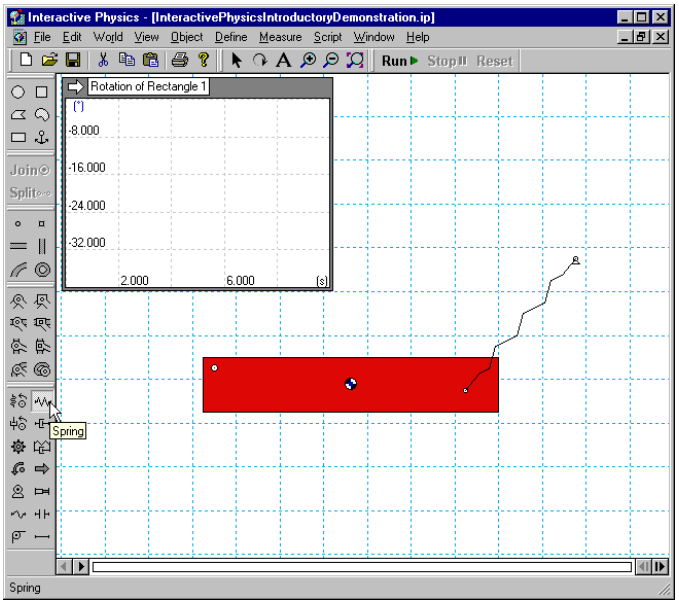
1.8 Adding Air Resistance

1. Under the World menu, select Air Resistance, click on Low Speed, slide the slider to the value 0.7 kg/(m * s), and click [OK].
2. Click Run and observe the exponentially decaying oscillations and notice that the pendulum's center of mass comes to rest directly below the pin. Click Reset.



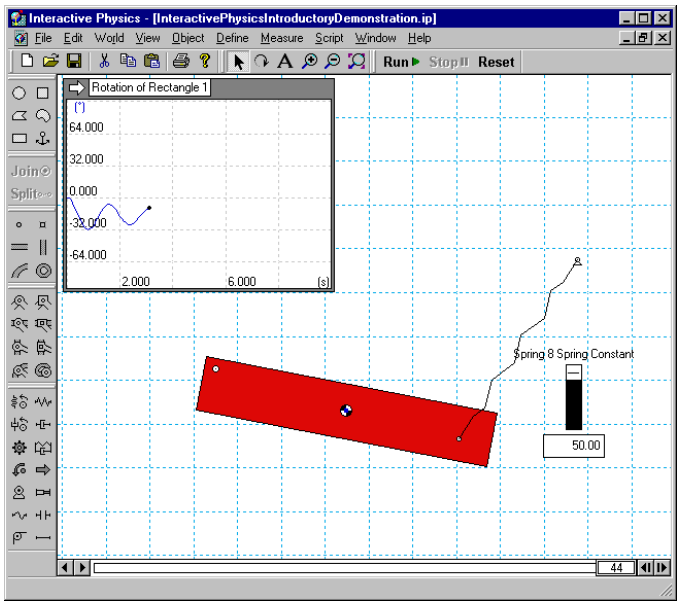
1.9 Adding a Spring

1. To add a spring, click on the Spring tool. Click on the upper right-hand corner of the block and stretch the spring up and to the right.
2. Click Run and observe the pendulum's higher natural frequency and new equilibrium position. Click Reset.



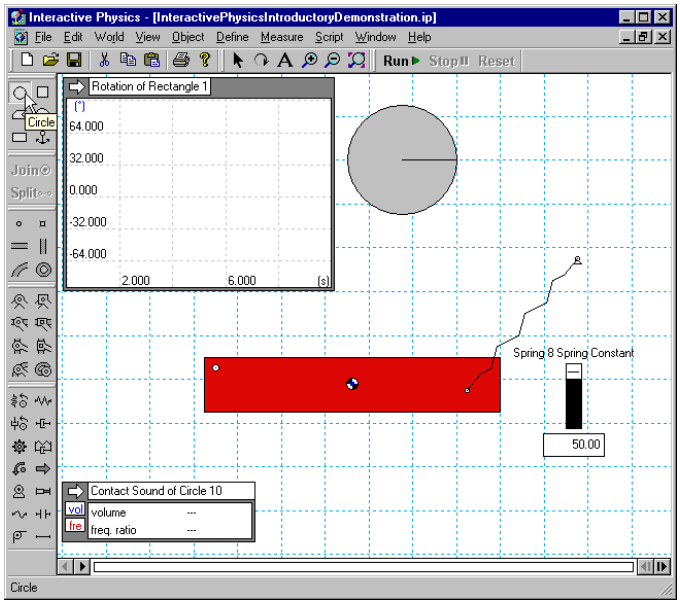
1.10 Controlling the Spring Constant

1. To control the spring constant, select the spring. Under the Define menu, select New Control, then select Spring Constant.
2. The slider that controls the spring will appear in the left-hand side of the workspace. To move the slider's location closer to the spring, click on its title and drag it next to the spring.
3. To see the effect of varying the spring constant, click Run and observe that the equilibrium angle of the pendulum is a function of the spring constant (move the slider up and down while the simulation is running). Click Reset.



1.11 Collisions with a Circle

1. To create a circle, click on the Circle tool, then click in the workspace and draw a circle. (If your rectangle is too high on the screen, press CTRL-A and drag all the objects lower in the screen.)
2. Click Run to start the simulation and observe that the circle bounces and rolls on top of the rectangle. Automatic collision and contact is a very useful feature in Interactive Physics (even the elastic and frictional properties of objects may be varied). Click Reset.



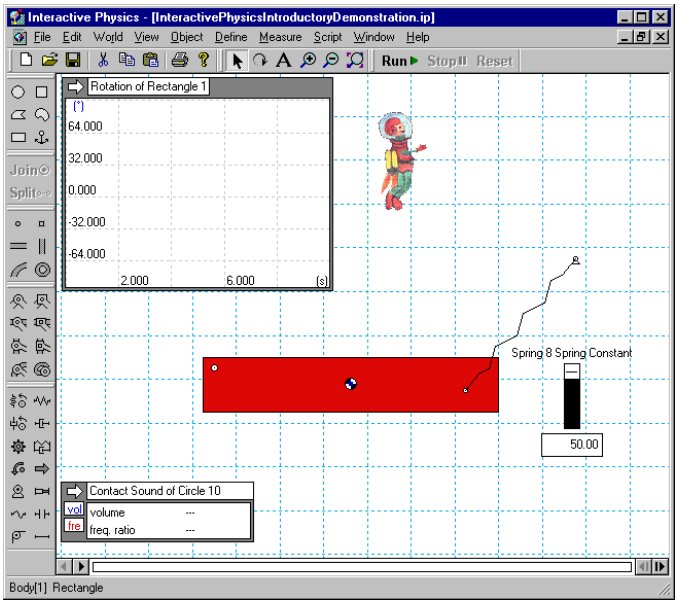
1.12 Attaching a Picture to an Object

1. Using the Windows Explorer, go to the directory where Interactive Physics is installed, for example, D:\Program Files\IP 2000. Navigate to the sub-directory called Picture Library and then its sub-directory People.
2. Double-click on the bitmap file "Spaceman.bmp." This should open the bitmap in a program such as Paint.
3. In Paint, select Select All from the Edit menu to select the entire bitmap. Select Copy from the Edit menu to copy the bitmap to the Clipboard.
4. Go back to Interactive Physics.
5. Select Paste from the Edit menu to paste the spaceman image from the clip-board to the Interactive Physics workspace.
6. To attach the spaceman bitmap to the circle, click and select the spaceman. Then hold down [Shift] while you click and select the circle.

Note: Interactive Physics was designed to be easy-to-use. In this exercise, the only time you need to touch the keyboard is to hold down the [Shift] key.

7. Select Attach Picture under the Object menu. Notice that the circle object has disappeared and has been replaced by the spaceman image.

8. **Click Run to run the simulation. Click Reset.**

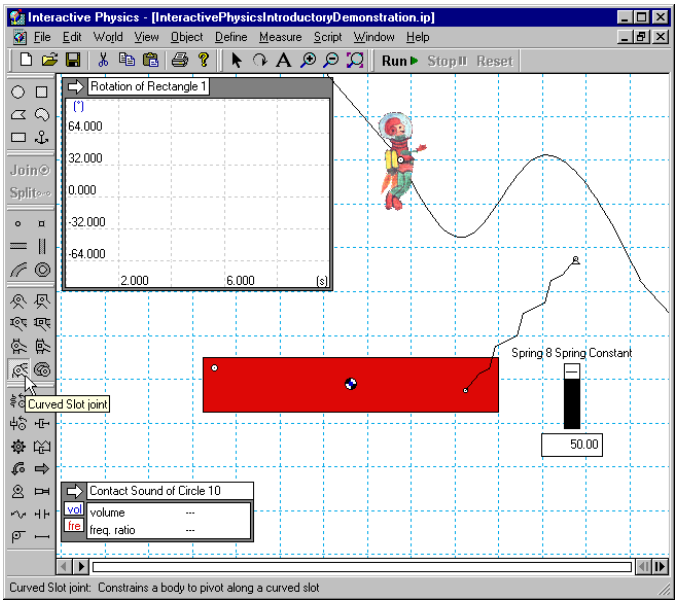


1.13 Adding Sound

1. To add sound, click on the spaceman and select Contact from the Measure menu.
2. Click Run to start the simulation to hear the sound when the spaceman contacts the block. Click Reset.

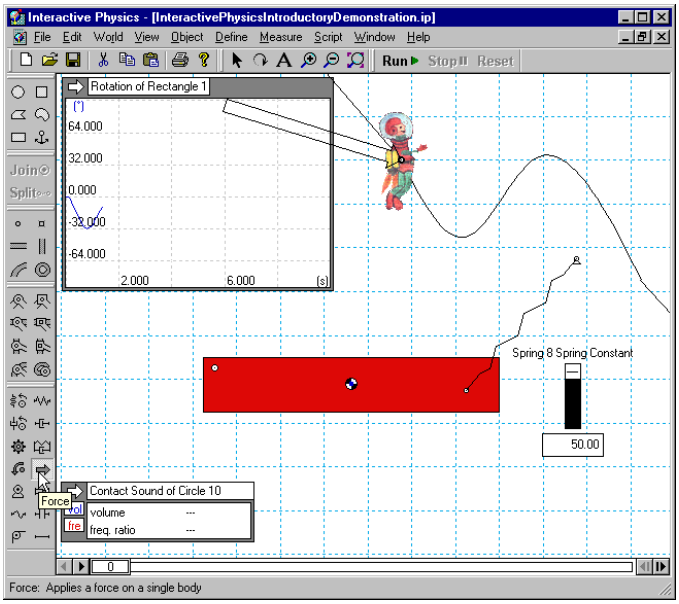
1.14 Adding a Curved Slot Joint

1. To add a Curved Slot Joint, click on the Curved Slot joint tool.
2. Click on the spaceman and then click on a couple of other places to the right of the spaceman, and then double-click to complete the slot.
3. Click Run to start the simulation and observe that the spaceman slides down the curved slot. Click Reset.



1.15 Adding a Force

1. To add thrust to the spaceman to overcome air resistance, click on the Force tool, then click on the spaceman, then move the mouse to the left and click again.
2. Click Run to start the simulation and observe that the spaceman is overcoming air resistance and is moving more quickly down the curved slot. Click Reset.



1.16 Running Demo Files

1. Under the Script menu, click on "Run All Demo Files".
2. Sit back and enjoy a series of demos on a variety of physics topics.

